STUDENT LIFESTYLE PERFORMANCE PREDICTOR

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INTRODUCTION

From my point of view as a student, I have seen times when small things in life like how time I spend studying how well I sleep and how much time I spend on my phone affect how well I do in class. Sometimes I study a lot. I still do not do well. The thing is, it is not always clear which of these things will help or hurt my performance in class.

For example, I might study for a time but if I do not sleep well I might still do poorly in class. This made me wonder: can we find a way to balance all these things like studying and sleeping and phone use and get some insights into how they affect my performance in class.

For this project I made a tool that uses machine learning to predict how well a student's lifestyle is doing. My goal was to create a system that can predict how a student, like me will do in class. The system will say if the student will do Poor, Average or Good in class. I based this on things, like how hours the student studies, sleeps and uses their phone. I wanted to see if the tool can really predict how well a student will do in class based on these things like study time and sleep and phone use.

PROBLEM STATEMENT

During my time in school I saw students, including myself have a hard time balancing school work with everyday things we do. Sometimes we try hard but still do not get the results we want while small things like not sleeping enough or spending too much time on computers and phones can quietly hurt our school work. These daily habits often seem like no deal but over time they can make a big difference, in our school work.

One thing I learned is that most of us do not really understand how the things we do every day affect our school work. We usually just. Think we know what is going wrong instead of really looking at it. There is not a way for students to look at their daily habits and clearly see how they affect our school work.

That is why I decided to make a system that looks at things we can measure and predicts how well we will do in school in a simple way that is easy to understand. I did not just want it to give us an answer but to help students learn how they can get better one step at a time.

I think this is a problem because students today have many things that distract them from school work. A tool that gives us feedback based on the things we actually do every day can help us make choices and slowly get better at our school work.

OBJECTIVES

My project was about building a Machine Learning model to figure out how well students do in school based on the things they do every day. I wanted the Machine Learning model to put students into three groups: Poor, Average and Good so it is easy to see how they are doing.

I also wanted to make the Machine Learning model something people can actually use. Of just using information that is already there I made it so users can put in their own information and get answers that are just for them. This makes the project more useful for students who want to see how they are doing. The Machine Learning model is really helpful because students can use it to get an idea of their own progress.

I wanted the Machine Learning model to help students get better too. If the Machine Learning model says a student is not doing well it gives them some ideas to get better like studying sleeping better using their phone less or going to school more. The Machine Learning model does not just say how a student is doing it also helps them get better. This is really important because it means the Machine Learning model is not a tool but something that can really make a difference in a student's life.

While I was working on this project, I also learned about the basics of Machine Learning like how it works and how it makes guesses based on the information it gets. It was really cool to see how the ideas I learned about work in life with the Machine Learning model. I got to see the Machine Learning model, in action and understand how it uses the information it gets to make decisions.

Overall, I did not just want to make a Machine Learning model I wanted to make something that can really help students develop habits and do better in school with the Machine Learning model. The Machine Learning model is a tool that can help students achieve their goals and become learners.

METHODOLOGY

To finish this project, I did things one step at a time. This helped me understand each part of building a Machine Learning model. I made a list of things like how many hours I study, how long I sleep how often I go to class how much time I spend looking at screens and how much I exercise. I gave each piece of information a label that said if my performance was Poor, Average or Good. I did this based on what I thought was real.

When my data was ready, I put it in order. I separated it into two parts: the information I put in and the results I got. This made it easier to use the data. Then I split my data into two groups: one to teach the Machine Learning model and one to test the Machine Learning model. This way the Machine Learning model could learn from some of the data. Then see how it did with data it had not seen before.

I picked the Decision Tree algorithm because it is easy to use and understand and it works well with sets of data like mine. The Decision Tree algorithm also helped me see how the Machine Learning model made decisions at each step.

After I taught the Machine Learning model, I added a feature that lets users put in their information and get predictions. This is really cool because it lets people see what they can do to improve. The Machine Learning model is really helpful. Finally, I made a plan to help people improve which makes the project more useful and focused on the people who will use the Machine Learning model. I included this plan because I wanted to make the Machine Learning model really helpful, to people who will use it.

SYSTEM DESIGN

When I was creating the system, I wanted to make sure it was simple to use and understand. I didn't want people to get confused. I divided the system into three parts:

- The part where you input information
- The part where the system processes that information
- The part where the system shows you the results.

This made the whole process very organized and easy to follow.

In the part where you input information you need to tell the system about your life. This includes things like

- How hours you study?
- How long you sleep?
- How often you attend classes?
- How time you spend on screens?
- How much you exercise?

I made sure it's easy to give the system this information. You don't need to know anything about computers.

The system then takes this information. Changes it into a format that the Machine Learning model can understand. It gives this information to the model, which analyses it and tries to predict your performance based on what its learned. The Machine Learning model is, like a computer program that can learn and improve over time.

The part where the system shows you the results tells you how well you'll do in a way that’s easy to understand. If you're not doing well it also provides a plan to help you improve. This plan has suggestions that you can use to boost your performance. The system is designed to be helpful and user-friendly. I aimed to make it useful not complicated.

KEY DECISIONS

When I was working on this project I had to make some choices to keep the system easy to use and understand. One of the things I decided was to use the Decision Tree Classifier for the Machine Learning part. I picked this one because it is simple and does not need a lot of work to get the data ready. It also shows clearly how the system makes decisions based on what you put in. Since this project is for people who are just starting out I wanted to make sure it was easy to understand.

Another important thing I decided was to make a set of data by myself. It is not huge. It gave me complete control over what was in it and helped me see how the system learns from it. I only included things that really affect how well students do like how time they spend studying, sleeping and going to class.

I also wanted to add a part that helps students get better. I did not just want the project to guess how they would do I wanted it to be helpful. So I added a feature that gives students ideas for improvement. This makes the system more useful for students.

Letting users interact with the system by putting in their information was another big choice. This way users can see how the results relate to their lives.

Overall, I tried to make the system clear, simple and useful when I was making these choices. I wanted the Decision Tree Classifier and the whole project to be easy to understand and helpful, for students.

CHALLENGES FACED

During this project I faced challenges. They helped me learn a lot. One big problem was the dataset. There were not data points. The model did not always give predictions. It also did not work well in all situations. I realized that bigger and more diverse datasets are important in Machine Learning. Machine Learning needs datasets.

Another challenge was matching the input format during prediction. It had to match the format used while training the model. Small mistakes could lead to warnings or wrong outputs. I had to be careful with the data. I had to structure it.

At first the Decision Tree algorithm was confusing. It took time and practice to understand it. The model splits data based on features. It makes decisions based on these features.

I also designed improvement suggestions. I wanted them to be simple and helpful, for students. I wanted them to be realistic.

I worked on handling input errors. This made the program easier to use. These challenges helped me improve my understanding. They also helped me improve my problem-solving skills.

CODE

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
data = {
    'study': [2, 5, 6, 1, 7, 3, 4, 8],
    'sleep': [5, 7, 6, 4, 8, 6, 7, 7],
    'attendance': [60, 80, 90, 50, 95, 70, 85, 92],
    'screen': [6, 3, 2, 7, 2, 5, 4, 1],
    'exercise': [0, 1, 1, 0, 1, 0.5, 1, 1],
    'performance': ['Poor', 'Average', 'Good', 'Poor', 'Good', 'Average', 'Average', 'Good']}

df = pd.DataFrame(data)
X = df[['study', 'sleep', 'attendance', 'screen', 'exercise']]
y = df['performance']


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = DecisionTreeClassifier()
model.fit(X_train, y_train)

def improvement_plan(study, sleep, attendance, screen, exercise):
    print("\nImprovement Plan:")

    if study < 4:
        print("- Increase study time to at least 4-6 hours daily")
    else:
        print("- Maintain your study consistency")

    if sleep < 6:
        print("- Get at least 6-8 hours of sleep")
    else:
        print("- Keep maintaining a healthy sleep routine")

    if attendance < 75:
        print("- Improve your attendance to above 75%")
    else:
        print("- Good attendance, keep it up")

    if screen > 4:
        print("- Reduce screen time to below 3-4 hours")
    else:
        print("- Screen time is well managed")

    if exercise < 1:
        print(" - keep trying to maintain 30-60 minutes of exercise daily")
    else:
        print("- Good job")

    print("- Stay consistent and follow a disciplined routine")


#user input
print("\nEnter student details:")

try:
    study = float(input("Study Hours: "))
    sleep = float(input("Sleep Hours: "))
    attendance = float(input("Attendance (%): "))
    screen = float(input("Screen Time (hours): "))
    exercise = float(input("Exercise Hours: "))
except ValueError:
    print("Invalid input! Enter numbers only.")
    exit()

input_data = pd.DataFrame([[study, sleep, attendance, screen, exercise]],
                          columns=['study', 'sleep', 'attendance', 'screen', 'exercise'])

prediction = model.predict(input_data)[0]
print("\nPredicted Performance:", prediction)

if prediction in ["Poor", "Average"]:
    improvement_plan(study, sleep, attendance, screen, exercise)
else:
    print("\nGreat! Keep maintaining your current routine.")
```

RESULTS

```
Enter student details:
Study Hours: 6
Sleep Hours: 5
Attendance (%): 92
Screen Time (hours): 3
Exercise Hours: 1

Predicted Performance: Good

Great! Keep maintaining your current routine.
```

The final system works as I expected. It is able to generate predictions based on the user input. When I enter values such as study hours, sleep duration, attendance, screen time and exercise the model analyses these user inputs. Predicts a performance category. Poor, Average or Good. Based on what the model learned during training.

While I was testing the system I noticed some patterns in the results. For example, better habits generally lead to predictions. If I enter inputs with study hours, consistent sleep and good attendance the model often results in a Good performance category. The model does this with the study hours, sleep duration, attendance, screen time and exercise that I enter. On the hand if I enter lower study time or excessive screen time the model tends to produce Average or Poor predictions. This shows that the model is able to reflect relationships between lifestyle habits and academic performance like the study hours, sleep duration, attendance, screen time and exercise.

The improvement plan feature also works effectively by providing practical suggestions whenever the predicted performance is not satisfactory. This makes the system more useful than just a system that predicts

The model is quite simple. It successfully demonstrates how Machine Learning can be applied to real life situations like the study hours, sleep duration, attendance, screen time and exercise. Overall the results show that even a basic model can generate helpful insights when designed thoughtfully using the user input, like study hours, sleep duration, attendance, screen time and exercise.

LIMITATIONS

This project has a problem that I found out about when I was working on it. The biggest problem is that I used a group of information to teach the Machine Learning model. This means the model does not always make predictions and it does not really show what happens in the real world. I learned that it is really important to have a lot of information when you are working with Machine Learning.

Another problem is that the model only looks at a things that people do every day like how much time they spend studying, sleeping, going to class using screens and exercising. These things are important. There are many other things that can affect how well someone does in school like how stressed they are if they are happy what they eat and how they study. The model does not look at these things.

I also used a way to make the model so it might not understand complicated relationships between different things.. The system only works on the command line right now which can be hard to use for some people.

The suggestions the system gives to help people get better are not very detailed. Do not really fit each person. Overall, these problems show that I can definitely make this project better, in the future. The Machine Learning model needs work to make it really good.

CONCLUSION

This project really helped me understand how Machine Learning works in a way. I learned that Machine Learning can be used for things that happen in my life as a student. For example, Machine Learning can look at how I study, how I sleep and how time I spend on screens to figure out how well I will do in school.

I saw that even a basic Machine Learning model can give me information if the data is put together and used in the right way. What I liked the most about this project is that it does not just predict how I will do in school. The Machine Learning system also gives me ideas on how I can improve, which makes it really useful for students like me. It does not just give me a result it actually helps me think about how I can do and develop good habits.

Course there are some things that we could do better like getting more data and looking at more things that can affect my school performance. I think this project is a great start. We can make it better. Add more to it over time.

I really liked working on this project. It helped me take the things I learned about Machine Learning, in class and use them to do something meaningful. Machine Learning is really cool because it can help me understand my life as a student and give me ways to make it better.

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